

Research Article

Effect of Static Rotor Eccentricity on End Winding Forces and Vibration Wearing

Hong-Chun Jiang¹,^{ID} Gui-Ji Tang¹,^{ID} Yu-Ling He¹,^{ID} Kai Sun¹,^{ID} Wei-Jun Li,²
and Lun Cheng³

¹Department of Mechanical Engineering, and the Hebei Key Laboratory of Electric Machinery and Failure Prevention, North China Electric Power University, Baoding 071003, China

²State Grid Zhejiang Electric Power Research Institute, Hangzhou 310014, China

³State Grid Hebei Electric Power Supply Co., Ltd., Shijiazhuang 050022, China

Correspondence should be addressed to Yu-Ling He; heyuling1@163.com

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In order to study the vibration wearing regularity and the strength failure point of stator end windings before and after static rotor eccentricity, the three-dimensional electromagnetic forces and the subsequent mechanical responses are studied in this paper. The electromagnetic force, stress, and deformation on the end winding of the QFSN-600-2YHG turbo-generator are calculated by the finite element method (FEM) through an electromagnetic-structure coupling. The radial vibration characteristics of the winding are verified by experiments. It shows that the vibration wearing in the same layer is more serious than that between two neighboring layers. For different layers, the interphase coils endure a larger wearing risk than the innerphase coils. Inside the same phase, the last coil along the rotating direction has the highest risk of insulation wearing. The occurrence of static rotor eccentricity will significantly increase the electromagnetic forces and the vibration amplitudes on some coils. The end-phase coil which is close to the minimum air-gap point is the most dangerous one due to the lasting over stresses and the intensified deformations.

1. Introduction

The stator windings are the key components which implement the energy conversion. As the capacity increases, the stator windings in generators, especially in large turbo-generators, will afford larger electromagnetic forces. Such electromagnetic forces will cause the winding to vibrate [1] and endure stresses [2, 3].

Scholars have paid much attention to the electromagnetic forces/stresses on end windings. Chow et al. used the integral-equation method to determine the magnetic field and the electromagnetic forces on conductors [4]. Khan et al. set up the force analysis model based on the image current analysis [5]. The key point of these two methods is to use Biot-Savart's law to calculate the magnetic field on the end region based on the integral performance. Albanese et al. further confirmed the effectiveness of such a method

[6]. The advantage of such a method is that it is able to obtain the magnetic flux density at an arbitrary point. Afterwards, the finite element method (FEM) was fast developed and more and more researchers employed FEM to study the electromagnetic forces. For instance, Kim et al. analyzed the distribution of the electromagnetic force on the end windings based on the Biot-Savart law and 3D FEM [7], while Fang et al. carry out the calculation of the stress distribution on end windings in a large water filling submersible motor during both steady state and transient operations [8, 9]. Also, Salon et al. presented the winding force variation under the transient operation in larger turbine generators [10]. Besides the running conditions (steady state and transient case), it is also found that some other factors such as the speed, the power factor, the saturation of the stator teeth, the magnetic field distribution, the winding connection, and the shape of the salient poles will also affect the force magnitude

TABLE 1: Key parameters of the QFSN-600-2YHG turbo-generator.

Parameter	Value	Parameter	Value
Rated capacity	667 MVA	Rated exciting current	4128A
Rated voltage	20000 v	Number of stator coils in series per phase	7
Rated rotating speed	3000 rpm	Number of turns in per rotor slot	8 (6 for slot 1 and slot 16)
Number of pole pairs	1	Length of rotor	6250 mm
Connection mode of stator winding	2Y	Stack length	6300 mm
Pitch	17	Inner diameter of stator	1316 mm
Power factor	0.9	Outer diameter of stator	2674 mm
Operating temperature	75°C	Inner diameter of rotor	500 mm
Indexing slots of rotor	50	Outer diameter of rotor	1130 mm
Real slots of rotor	32	Air-gap length	93 mm
Number of stator slots	42	Stacking factor	0.95
Rotor slot dimensions	60 * 40	Stator slot dimensions	160 * 70

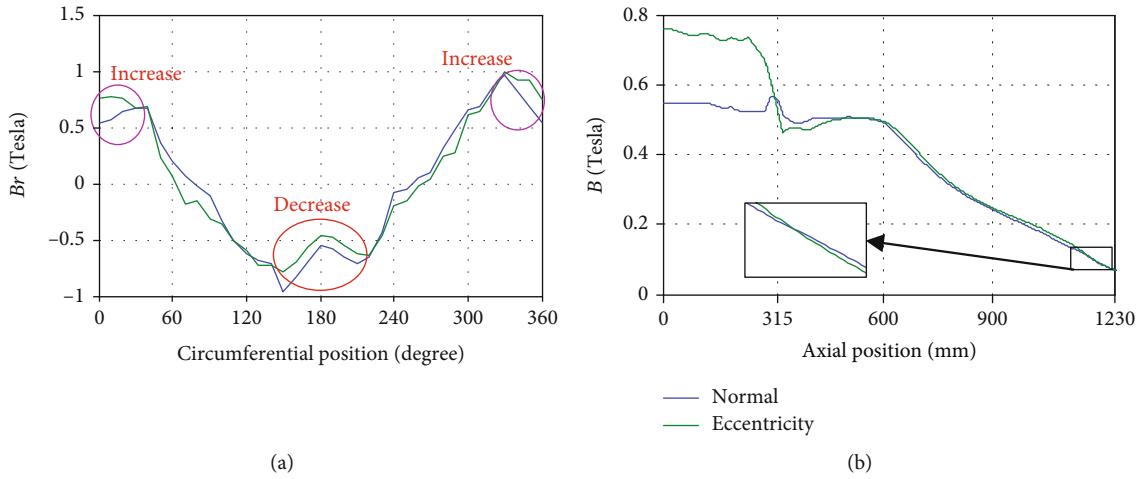


FIGURE 3: MFD variation: (a) circumferential distribution and (b) axial distribution.

3. Finite Element Analysis

3.1. Model Setup. In this paper, we take the QFSN-600-2YHG type turbo-generator as the study object, and the 3D transient solution type is employed for FEA. During the calculation, we establish the same physical model and external coupling circuit as Ref. [1]. The static rotor eccentricity is set along the X axis, and the eccentricity rate is 0.03.

The key parameters of the generator are listed in Table 1. The axial length of the stator core is 6300 mm which requests many computing resources in 3D calculation. To balance the calculation accuracy and the resource, one-tenth of the stator, namely, 630 mm, is set for the finite element analysis [1]. The stator core has 42 slots, with the detailed slot distribution illustrated in Figure 2(a). Each slot has two layers of coils, and each coil has an upper line and a lower line inside these slots. For instance, the 34th coil has the upper line in slot 34 and the lower line in slot 9. The windings are of double Y connected, and the winding distribution of phase A is illustrated in Figure 2(b).

3.2. Electromagnetic Force. The radial magnetic flux density variation is illustrated in Figure 3(a). It shows that the mag-

netic flux density curves display as a cosine function. As the eccentricity occurs, the MFD value near the 0 degree (X axis) will be increased, while near the 180 degrees, the MFD will be decreased. Such result well accords with Equation (7).

The MFD distribution in the axial direction is indicated in Figure 3(b), where the coordinate origin is set at the central point of the stator core. Since half of the stator length is 315 mm, the MFD will obtain the larger values in [0, 315], while outside the stator core, the MFD amplitude will be decreased. The occurrence of the static rotor eccentricity will generally increase the MFD inside the stator core, namely, the 0-315 mm region.

To better analyze the electromagnetic force properties, the force of the line part and the force on the end part are calculated individually. The force density is firstly calculated. Then, the further integral operation is carried out via the field calculator to obtain the radial, the tangential, and the axial force components, respectively. The positive directions for the radial force and the axial force are set along the radial and the axial axis, respectively, both from inside to outside. For the tangential force, the positive direction is set along the forward rotating orientation. According to the aforementioned theoretical result, the electromagnetic force of the coil